KODA

43 EUR per MWh out of balance, nobody knows which way

43 €/MWh

average cost of one MWh of imbalance

mean |imbalance − spot|, validation

25 %

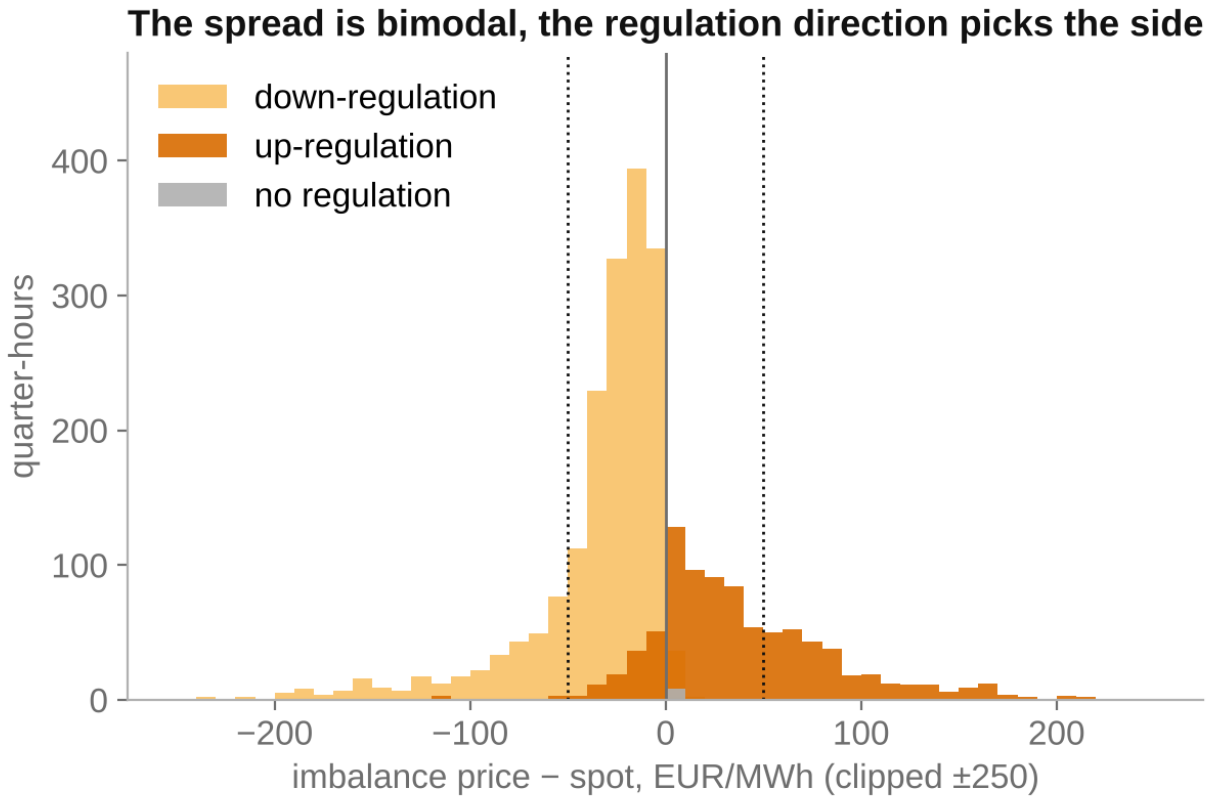
of quarter-hours beyond ±50 EUR/MWh from spot

two price modes, up and down

10 h

ahead a plant must plan, with a full-hour origin

no public forecast of this price exists



Turku Energia runs CHP and electric boilers: both can move power either way, if they know which way the price goes.

The market is a rule, physics is measurable, one model fails

The price is a rule, not a mystery

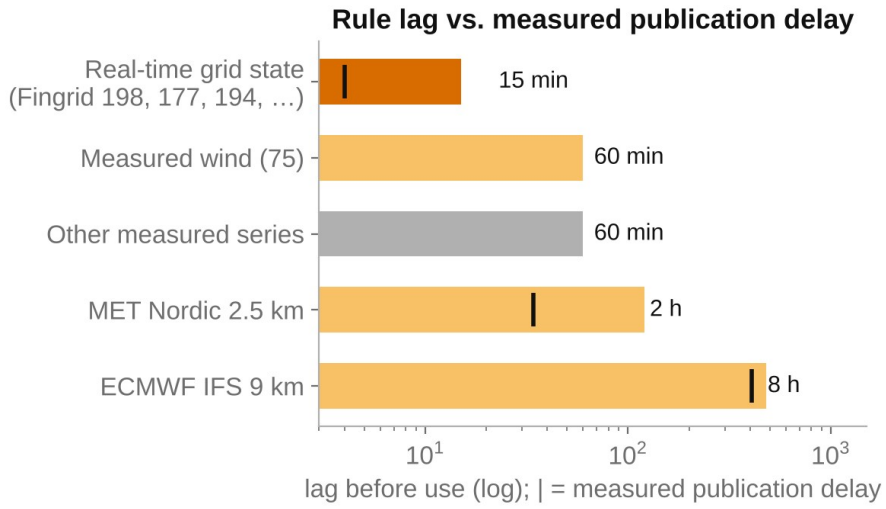
Fingrid's 1 June 2026 rule rebuilt from public series reproduces 99.6 % of quarter-hours exactly. Only the regulation direction and the balancing price need forecasting; spot is known, so we forecast the spread.

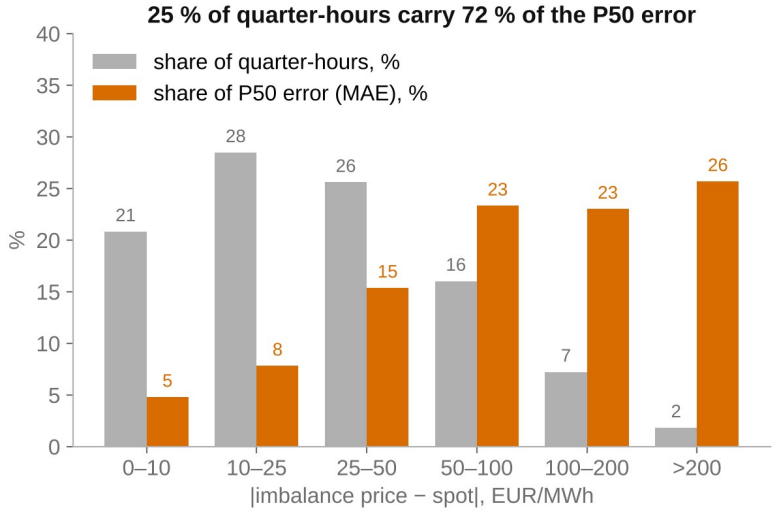
Measured weather beats modelled weather

Measured wind minus Fingrid's day-ahead forecast, persisted, beats the forecast itself at 0–2 h (235 vs 320 MW). The grid's own surplus, frequency and flows are public 1–4 minutes after each quarter-hour.

A smooth model cannot see a spike

One quantile model: MAE 26 on down-regulated quarter-hours, 77 on up-regulated. 72 % of the error sits in the 25 % of spiky quarter-hours, so the value is in probabilities, not in the point forecast.





Three LightGBM heads mixed into one distribution per member

Feature table

one row per (origin, step)
76–93 features
all read through the availability mask

market state · grid state (15 min) · measured wind − forecast · spot and day-ahead forecasts · NWP · cross-border · calendar

LightGBM classifier

P(up) · 110 trees · P(down), P(none) from it

7 LightGBM quantile heads

spread | up-regulation · q05…q95 · 150 trees · trained on up rows only

7 LightGBM quantile heads

spread | down-regulation · same, on down rows only

Mixture

heads widened ×2 around their medians, then

$F = P(up)\cdot F_{up} + P(down)\cdot F_{down} + P(none)\cdot 1[x \geq 0]$

$P10/P50/P90 = F^{-1}$
P(spike) = tail mass beyond ±50

Outputs per row

p10, p50, p90
p_up, p_down, p_none
p_spike_high, p_spike_low
top_drivers (TreeSHAP)
data_cutoff_utc

Explainable: TreeSHAP on every head attributes each forecast to its drivers, a mathematical account of a model that is otherwise a black box.

Full SHAP tables (GitHub)

× 2 lead blocks: steps 1–8 (with grid state) and 9–40

Then the quant layer

8 members

2 weather sources × {default fit, second seed, last-7-weeks window, asinh target}, averaged at the quantile level

Held-out widths

band factor per lead block and direction certainty, fitted on the other four weeks

Hedge

spike probabilities averaged 50/50 with the previous chain; Brier convexity makes it never worse than the mean

Bootstrap

every choice tested on 300 day-block resamples

Four things no smoothed price model does

1

Sees when the weather model is wrong

Physical measurements from the observation network, not just forecasts. Measured wind against the day-ahead forecast and the grid's own surplus, frequency and flows reach the model 15 min after the quarter-hour. When the regional NWP drifts, the error is a feature before it becomes a price.

235 vs 320 MW wind error at 0–2 h

2

Multi-head, one pass, built to be extended

Direction and both conditional prices are separate heads over one feature table, so inference is one pass and data is used three times. A plant-specific head (own output, boiler, storage) can be trained on top without touching the price model, and training continues as new weeks arrive.

8 members, ~100 s each to train

3

Self-healing when a feed drops

Trained with gaps: 9–16 % of grid-state quarter-hours are missing and NaN is a first-class value for the trees. Two independent weather sources, the latest complete value with its age as a feature, and lead blocks that fall back to slower data keep the forecast running when one or several sources go dark.

no single data source is load-bearing

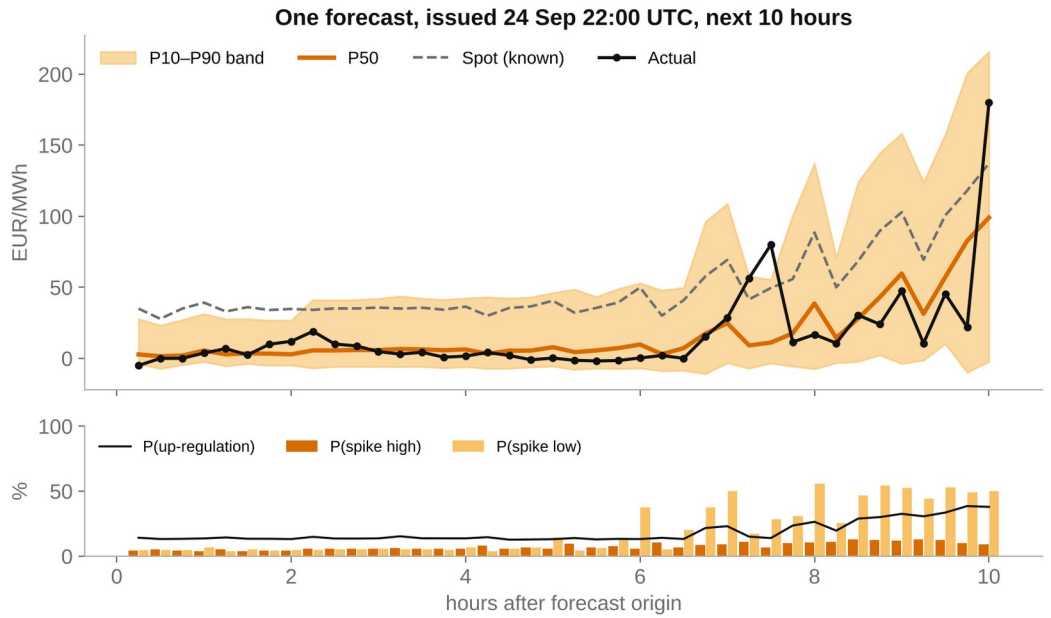
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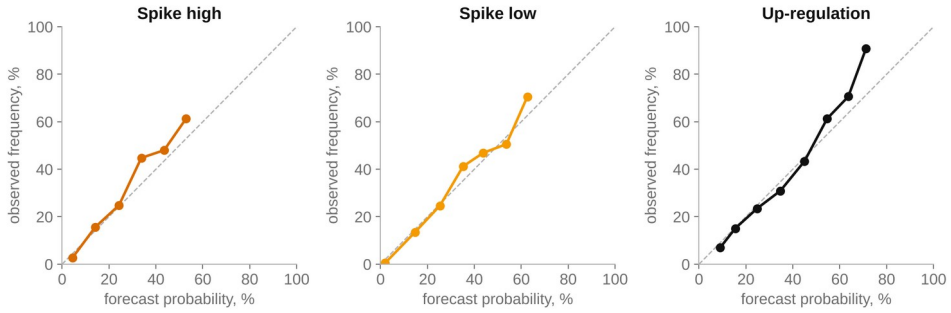
Quant-trading machinery on the distribution

Bagging over seeds, training windows and target transforms; a hedge between two model chains; widths fitted held out week by week; day-block bootstrap for every decision. The same discipline used on trading books, applied to a physical price.

better in 100 / 97 / 92 % of resamples

Beats the baseline on every score with honest probabilities





Our model vs. the organiser's baseline, validation 7 Sep – 4 Oct 2026, official score.py

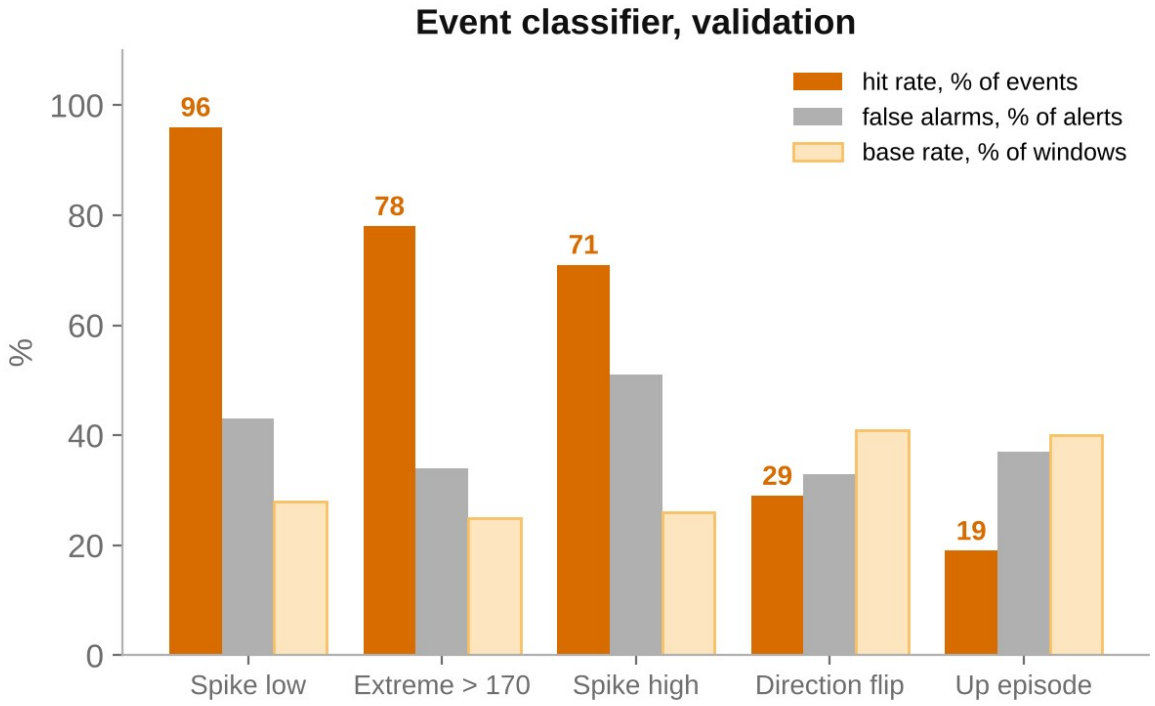
–38 %	**P50 error (MAE)** 66.2 → 41.0 EUR/MWh
–53 %	**band error (pinball P10/P90)** 24.2 → 11.3
82 %	**band coverage** target 80 %, baseline 68 %
–23 %	**spike Brier** 0.113 → 0.087
–15 %	**direction Brier** 0.451 → 0.385

Better in 100 % of day-block resamples on all four metrics. Published 15-min models beat persistence by 12–25 %; we beat it by 38 %.

A panel that points at the hours needing attention

16:00–18:00 · P(spike ↑) 34 %, 3× normal, confidence high
Drivers: wind forecast down, consumption above forecast, net import falling.
Check: CHP headroom, boiler load, heat storage.

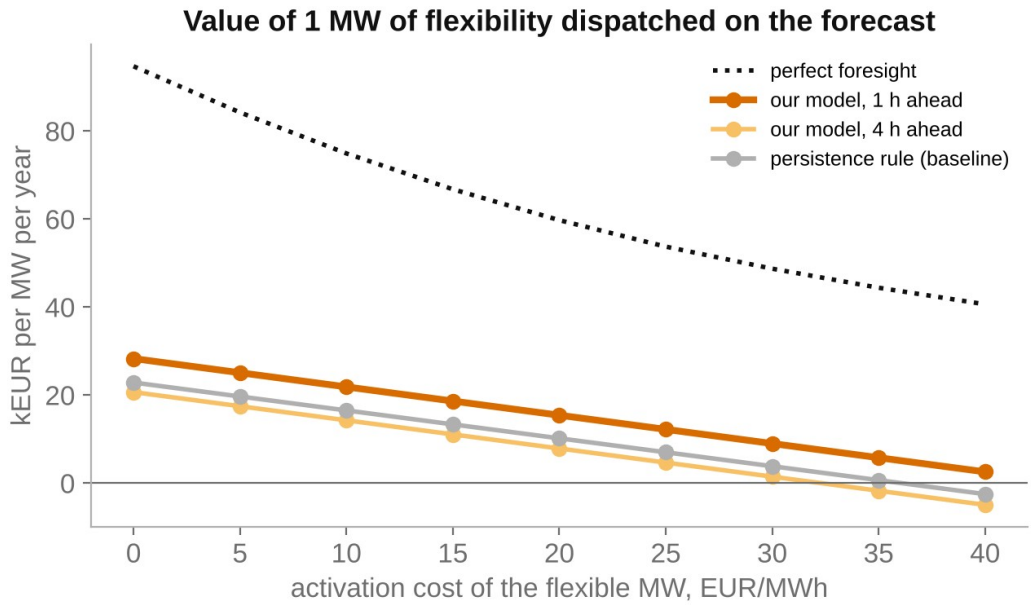
Price forecast	band, direction, spike risk, top drivers per quarter-hour
Rule alerts	forecast moved, band wide, two weather models disagree, data stale
Event classifier	P(spike episode, extreme price, direction flip) per 0–2 / 2–5 / 5–10 h window, with a Venn–Abers interval and a lift vs. normal
Two personas	trader: position and flips · operator: CHP + electric boilers, what to check



Per event, four candidates (logistic, LightGBM, MLP, a rule on the forecast's own probabilities) compete on validation Brier; the learned model ships only where it clearly wins. Weak events are hidden rather than shown.

Suggestions to review, not operating instructions. The classifier does not change forecast_backtest.csv.

Trains in minutes, transfers to any asset, keeps learning



Today on public data: 15 k€ per flexible MW per year, 50 % above the naive rule. Perfect foresight: 51 k€. The gap is what asset data and on-site observations can close.

~100 s

to train a member, one pass to infer

Retrain on every new week. Deploy the same pipeline on a new site or market by pointing it at its data.

One base

shared feature table, heads added on top

A plant-specific head (own CHP, boilers, storage, wind + BESS) trains on the same base without touching the price model.

On-site eyes

benefits from observation systems

Measured weather at the generation site, for example from WeatherSens AI's drone platform, enters as a feature the same way measured wind does today.

Keeps learning

outcomes become the reward

Every dispatch decision is scored against the realised spread; that signal retrains the heads and can drive reinforcement learning on top.

Value: tools/eval/economic_value.py on the validation forecasts; four weeks annualised, single-price settlement, plant costs not public.